Electrochemistry at ω-Hydroxy Thiol Coated Electrodes. 4. Comparison of the Double Layer at ω-Hydroxy Thiol and Alkanethiol Monolayer Coated Au Electrodes

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The diffuse-layer potential and potential of zero charge at alkanethiol and ω-hydroxy thiol monolayers adsorbed at Au electrodes are extracted from differential capacitance data and compared with the predictions of a two-capacitor model for the electrode double layer. Modeling the double layer as an electrolyte-independent monolayer capacitor in series with a diffuse-layer capacitor calculated from the Gouy–Chapman theory, close agreement is found between the calculated and measured diffuse-layer potential versus electrode potential curves. Heterogeneous electron-transfer rates for $Fe(CN)_6^{3-}$ measured at Au electrodes derivatized with dodecanethiol and 14-hydroxytetradecane-1-thiol monolayers are strongly dependent on the electrolyte concentration but are nearly independent of the electrolyte concentration once the data are corrected for the diffuse layer calculated via the two-capacitor model. Significant differences in the double-layer structure between the dodecanethiol and ω-hydroxy thiol monolayer coated Au electrodes are noted and linked to the large difference in the hydrophobicity of the two monolayer surfaces.

Introduction

The structure of the electric double layer present at an electrode/electrolyte interface can dramatically affect the rates of heterogeneous electron transfers. For electrode potentials positive and negative of the potential of zero charge, a portion of the applied potential is dropped outside of the plane of closest approach of nonspecifically adsorbed redox species within the diffuse layer. This positive or negative diffuse-layer potential, ϕ_2, distorts the concentrations of ionic species within the double layer from their bulk values. In addition, the electrode overpotential experienced by electroactive species at the outer Helmholz plane (OHP) is diminished from the applied overpotential by this diffuse-layer potential.

Heterogeneous electron-transfer data can be corrected for the effect of the diffuse-layer potential if it can be determined from electrocapillary curves or from differential capacitance measurements.[1] The specific adsorption of solution species onto the electrode surface and the possible potential dependence of the compact-layer capacitance often make determining the diffuse-layer potential a difficult and inexact proposal.[2] An alternate approach is to decrease the size of the diffuse-layer potential and thereby minimize the double-layer effect. Sears and Anson have proposed controlling the concentration of specifically adsorbed anions to balance the positive charge density of Hg electrodes positive of the point of zero charge.[3] Another more general approach is to allow a greater portion of the applied electrode potential to drop within the OHP by covering the electrode with a thin insulating film. In our labs, we have minimized the diffuse-layer potentials at Au electrodes through the spontaneous self-assembly of ω-hydroxy thiol monolayers.[4] Because of the low dielectric constant of these hydrocarbon monolayers, nearly all of the applied potential is dropped within the monolayer, resulting in significantly reduced diffuse-layer potentials.

Our main impetus for studying electron-transfer reactions at these insulated electrodes is that they enable us to characterize the redox properties of molecules at electrode potentials not accessible to uncoated electrodes. These monolayers limit the closest approach of redox molecules to the electrode surface, dramatically decreasing the rate of heterogeneous electron transfer.[5] Voltammograms measured at ω-hydroxy thiol monolayer coated electrodes can be analyzed to obtain the reorganization energies of electroactive molecules[4] and to probe the adiabaticity of the electron-transfer process.[6] A quantitative assessment of the double-layer structure of monolayer-coated electrodes is an important step in using these insulated electrodes because it allows one to calculate the absolute concentration of redox active species at the electrode surface from a knowledge of their concentrations in the bulk solution. Once the concentration profile of a redox species near the surface of an insulated electrode is known, one can extract absolute electron-transfer rates in the same manner that such data are obtained at thiol monolayers covalently derivatized with redox centers.[7]

Previously, we have presented a simple two-capacitor model for the double layer at these insulated electrodes.[5b] This model was used to estimate the small diffuse-layer potentials at the ω-hydroxy thiol monolayer coated Au electrodes. In this paper, we quantitatively test this model and extract the potential of zero charge from capacitance and electron-transfer measurements of ionic electroactive molecules as a function of the electrolyte concentration. This approach is extended to Au electrodes derivatized with self-assembled monolayers of alkanethiols, and a comparison of the double-layer structures at alkanethiol and ω-hydroxy thiol monolayer coated Au electrodes is made.

Experimental Section

The ω-hydroxyalkanethiols were synthesized from hydroxyalkyl bromides by treatment with thiourea, followed by NaOH and purification by column chromatography, as described in our earlier paper.[4] Alkanethiols were purchased from Aldrich and used as received. Au electrodes (fabricated via radio frequency (RF) sputtering onto microscope slides) were cleaned by successive exposures to chromic acid and aqueous HF just prior to their immersion into ethanolic thiol solutions, as described previously.[4] The Au electrodes were kept in the *ca.* 30 mM ω-hydroxy thiol solutions for at least 4 h and typically overnight before their use in voltammetric experiments.

$HFe(4,4'\text{-dimethyl, }2,2'\text{-bipyridyl})(CN)_4$ and $HFe(2,2'\text{-bipyridyl})(CN)_4$ were synthesized by the method of Schilt.[8] All other chemicals were reagent grade and used as received. Deionized H_2O (MilliQ system from Millipore) was used for all solutions and for rinsing the electrodes.

Electrochemical experiments were performed using a nonisothermal three-electrode cell arrangement held at 0.0 °C by a thermostated bath. All electrochemical measurements were

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performed using a BAS 100A electrochemical analyzer. Electrolyte solutions were purged with purified nitrogen and kept under a nitrogen blanket during all voltammetric measurements. All potentials are reported versus an SCE reference electrode.

Results and Discussion

One of the benefits of studying redox reactions with insulated is that double-layer effects are minimized and are more easily calculated. A simple two-capacitor model for the double layer for ω-hydroxy thiol monolayer coated electrodes can be used to calculate the diffuse-layer potential. The electrode double layer is considered to be composed of two regions: the hydroxy thiol monolayer, which is assumed to be an ideal capacitor, and the diffuse double layer, whose capacitance can be calculated by Gouy–Chapman theory.[2] This simple model is identical to the treatment of the double layer by Stern,[9] except that we replace the capacitance of the compact layer with the much smaller capacitance of the ω-hydroxy thiol monolayer. The electrode capacitance per cm², C_{total}, is then given by

$$\frac{1}{C_{total}} = \frac{1}{C_m} + \frac{1}{C_{gc}} \quad (1)$$

where C_m is the capacitance of the monolayer film per cm² and C_{gc} is the capacitance of diffuse layer per cm² (at 0.0 °C) calculated from

$$C_{gc} = 238.2zc^{1/2}\cosh(21.3z\phi_2) \quad (2)$$

where z is the absolute value of the charge of the z:z electrolyte and c is the concentration of the electrolyte in mol/L. The potential applied to the electrode, V (measured relative to the potential of zero charge), will be dropped within the two capacitors. Thus, the diffuse-layer potential, ϕ_2, can be calculated from

$$\phi_2 = V\frac{C_m}{C_{gc} + C_m} \quad (3)$$

$$\phi_2 = \frac{VC_m}{(238.2zc^{1/2}\cosh(21.3z\phi_2)) + C_m} \quad (4)$$

The value of ϕ_2 for a given electrode potential and electrolyte concentration can be found iteratively from this equation. The heterogeneous electron-transfer rate as a function of the electrode potential, $k(E)$, can be corrected for the double layer via the relationship

$$k(E)_{corr} = k(E)e^{-z\phi_2(E)/kT} \quad (5)$$

where z is the charge of the electroactive ion and kT is the product of Boltzmann's constant and the absolute temperature expressed in volts. Equation 5 corrects the heterogeneous electron-transfer rates only for the distortion of the concentration of the redox molecule at the OHP. The deviation in the applied overpotential with the effective overpotential for a redox molecule at the OHP is not very large for insulated electrodes and results in only a small correction to the voltammetric current compared to the correction given by eq 5.

The diffuse-layer potential calculated from the two-capacitor model above can be directly compared with that derived from the differential capacity of the monolayer-coated electrode. Figure 1 shows the differential capacitance of an Au electrode derivatized with a self-assembled $HO(CH_2)_{14}SH$ monolayer as a function of the concentration of the CF_3COONa electrolyte. These data were obtained from cyclic voltammograms measured using a scan rate of 500 mV/s. The differential capacitance was taken as the difference between voltammetric current measured during the positive and negative scans.[10] The distinct minimum in the differential capacitance measured at the lower electrolyte concentrations at *ca.* –0.07 V vs SCE marks the potential where the diffuse-layer capacitance is minimized at the potential of zero charge. As the concentration of the electrolyte is increased, the diffuse-layer capacitance increases, reducing the potential dependence of the total differential capacitance. Using the differential capacitance measured at the highest ionic strength as a measure of the monolayer capacitance, the diffuse-layer capacitance as a function of the applied potential can be calculated. From this diffuse-layer capacitance, the diffuse-layer potential can be calculated using eq 3. Figure 2 plots a comparison of the potential dependence of the diffuse-layer potential calculated from the data shown in Figure 1 (Figure 2A) with the potentials calculated using eq 4 (Figure 2B). Both the shape and magnitude of the diffuse-layer potential calculated from the capacitance data are in excellent agreement with those of the two-capacitor model presented above.

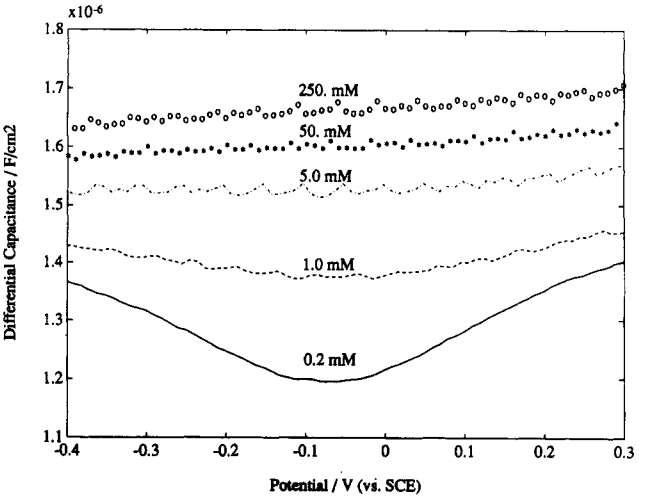


Figure 1. Differential capacitances measured at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode as a function of the concentration of the CF_3COONa electrolyte indicated in the figure. The capacitances were calculated from cyclic voltammograms obtained at 0.5 V/s scan rate at 0.0 °C.

As the ionic strength of the electrolyte is increased, the diffuse-layer potential becomes a linear function of the applied potential. The linearity of the diffuse-layer potential stems from the diffuse-layer capacitance becoming independent of the applied voltage. At electrolyte concentrations typically used for voltammetric experiments (0.1–1.0 M), the diffuse-layer potential is well described by

$$\phi_2 = \frac{C_m}{C_{gc} + C_m}(V_{app} - E_{pzc}) \quad (6)$$

where C_{gc} in μF/cm² (at 0.0 °C) is well approximated by

$$C_{gc} = 238.2c^{-1/2} \quad (7)$$

We observe some variability in the potential of zero charge measured by the position of the capacitance minimum from one electrode preparation to another. The potentials of zero charge (pzc) were observed to vary by approximately ±0.10 V. Electrodes coated with $HO(CH_2)_{11}SH$ gave the same pzc within experimental error. In addition to being independent of the length of the ω-hydroxy thiol monolayer, the pzc is also independent of the electrolyte. We have repeated these capacitance measurements in NaCl and $NaClO_4$ and extract the same potential of zero charge and a similar agreement between the measured and calculated diffuse-layer potentials. The value of the potential of zero charge measured at these ω-hydroxy thiol derivatized monolayers is in reasonable agreement with measurements of the pzc on bare Au electrode in nonspecifically adsorbing electrolytes.[11]

Kinetic Measurements. Because the presence of a diffuse-layer potential can dramatically distort the concentration of ionic redox species at the electrode surface, measurement of hetero-

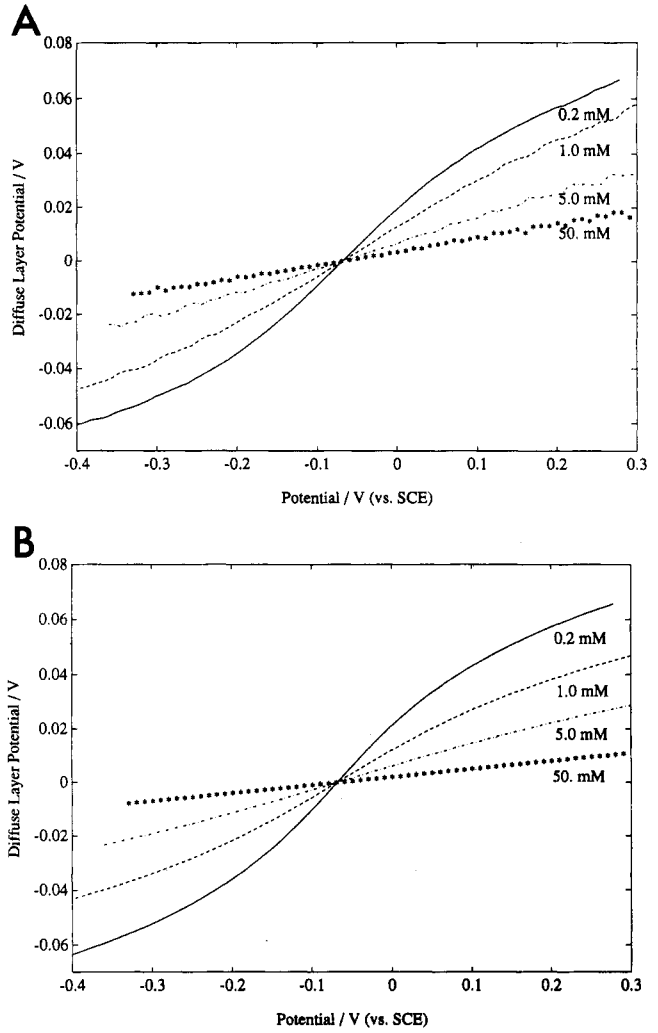


Figure 2. Electrolyte concentration dependence of the diffuse-layer potential versus the applied potential at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode. The diffuse-layer potentials shown in part A were calculated from the capacitance data shown in Figure 1 (see text). Part B plots the diffuse-layer potentials calculated using the two-capacitor model.

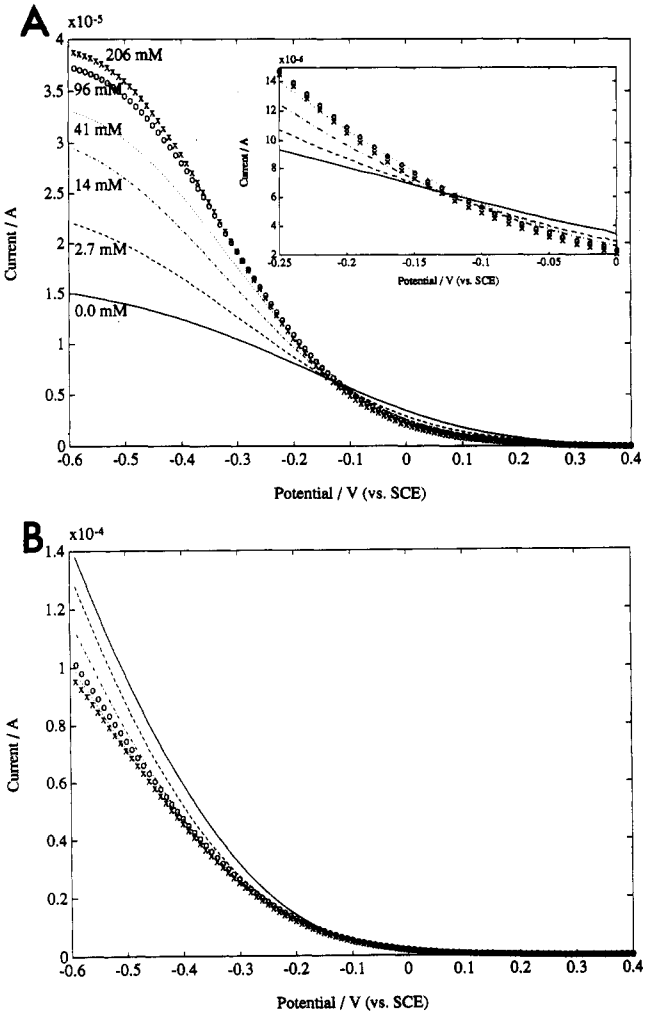


Figure 3. Correction of the heterogeneous electron-transfer rates for $Fe(dmbpy)(CN)_4^-$ reduction measured at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode using the two-capacitor model. Part A plots the reduction currents for 1.2 mM $Fe(dmbpy)(CN)_4^-$ measured at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode corrected for diffusion limitations via convolution techniques.[4] Part B plots these same currents after the diffuse-layer potential correction.

geneous electron-transfer rates as a function of the electrode potential and supporting electrolyte concentration can be used to characterize the double-layer structure of an electrode interface. De Levie *et al.*[12] and Anson *et al.*[13] have used this approach to determine the pzc and diffuse-layer potential for Hg electrodes. In these studies, the use of kinetically slow redox couples was essential in order to be able to measure heterogeneous electron-transfer rates over a significant range of electrode potentials. At ω-hydroxy thiol coated electrodes, the absolute electron-transfer rate can be controlled by controlling the thickness of the insulating monolayer film. This means that the electron-transfer rate for even the most kinetically facile redox species can be measured over a wide range of electrode potentials, allowing another independent test of the simple double-layer model presented above.

Figure 3a shows a series of voltammetric waves measured at a $HO(CH_2)_{14}SH$-derivatized Au electrode for the reduction of 1.2 mM $Fe(dmbpy)(CN)_4^{1-}$ as a function of the CF_3COONa supporting electrolyte concentration. These curves have been corrected via convolution techniques for the influence of diffusion limitations and so are proportional to the heterogeneous electron-transfer rate of this Fe complex at the insulated electrode.[4] As the electrode is polarized negative of the pzc, the increasing negative diffuse-layer potential repels anions from the OHP, decreasing the apparent heterogeneous electron-transfer rate. An estimate of the pzc for the electrode can be made from these data by noting the potential at which the electron-transfer rate is independent of the electrolyte concentration. This occurs at a potential of *ca.* –0.12 V vs SCE (see the inset plot in Figure 3A), which is identical to the pzc determined from the capacitance minimum for this electrode.

These voltammetric data can be corrected for the effect of the diffuse-layer potential using the two-capacitor model described above. The resulting corrected voltammetric curves (Figure 3B) are quite close to each other. The corrected current measured in the absence of supporting electrolyte is approximately 40% higher than the one measured at the highest electrolyte concentration, suggesting a small overestimation of the diffuse-layer potential. Alternate explanations for the small deviation between the curves in Figure 3B involve the uncertainty in the electrolyte concentration for the experiments performed at low ionic strength due to leakage of the electrolyte from the SCE reference electrode and the uncertainty in the diffusion correction.

Part of the problem in using this complex to probe the diffuse-layer potential is that it is singly charged and so is less susceptible to the ionic repulsion or preconcentration caused by the diffuse-layer potential. Figure 4A shows a series of voltammetric waves measured at a $HO(CH_2)_{14}SH$-derivatized Au electrode for the reduction of 2.0 mM $Fe(CN)_6^{3-}$ as a function of the CF_3COONa supporting electrolyte concentration. These curves have been corrected via convolution techniques for the small influence of diffusion limitations and so are proportional to the heterogeneous

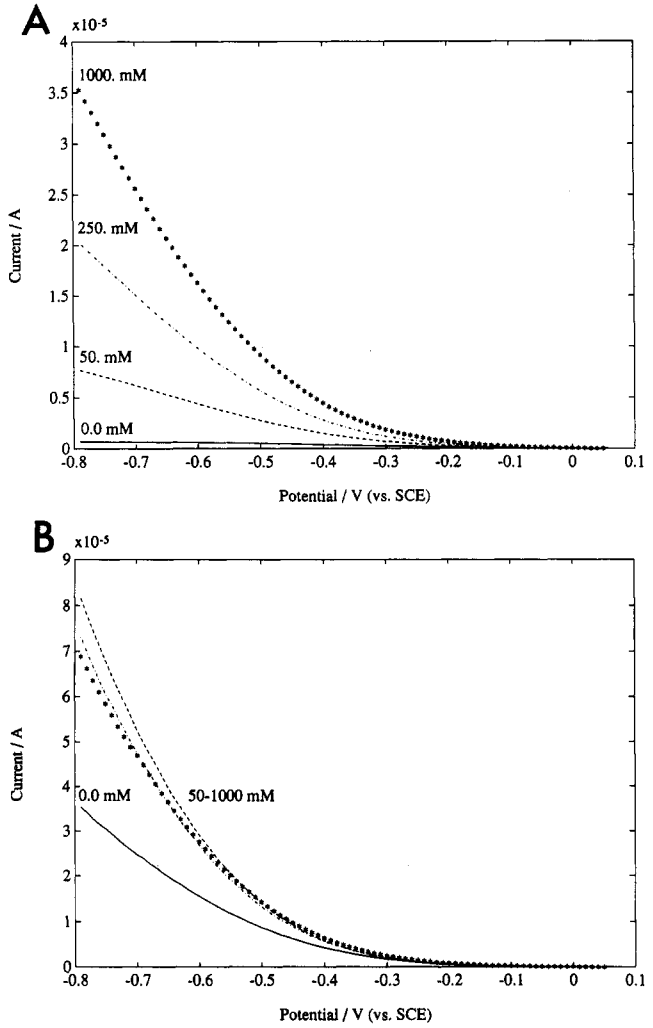


Figure 4. Correction of the heterogeneous electron-transfer rates for $Fe(CN)_6^{3-}$ reduction measured at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode using the two-capacitor model. Part A plots the reduction currents for 2.0 mM $Fe(CN)_6^{3-}$ measured at a $HO(CH_2)_{14}SH$ monolayer coated Au electrode corrected for diffusion limitations via convolution techniques.[4] Part B plots these same currents after the diffuse-layer potential correction.

electron-transfer rate of the ferricyanide at the insulated electrode.[4] Because of the high charge of the ferricyanide ion, the ionic strength dependence of the reduction kinetics at the most negative potentials is approximately 10 times larger than that observed for the $Fe(dmbpy)(CN)_4^{1-}$ monoanion.

Figure 4B shows this voltammetric data after they are corrected for the diffuse-layer potential calculated from the two-capacitor model. Except for the data obtained with no added supporting electrolyte, the voltammetric currents corrected for the diffuse-layer potential are nearly the same. The correction for the data obtained with no supporting electrolyte appears qualitatively worse for two reasons. First, the correction proposed above is only valid for a *z*:*z* electrolyte, and so one might expect some deviation in the diffuse-layer structure in the presence of 2 mM $Fe(CN)_6^{3-}$. In the calculation of the diffuse-layer potential, the ionic strength of the 2 mM $K_3Fe(CN)_6$ solution was used as the effective concentration of the *z*:*z* electrolyte.[14] Second, the correction required for the voltammetric curve without supporting electrolyte is considerably larger than that for the other curves. The uncorrected current measured at *ca.* −0.8 V in 1.0 M CF_3COONa is approximately 50 times larger than that measured in the absence of the supporting electrolyte. The same ratio of the corrected currents in Figure 4B is approximately 2. This translates into less than a 20% error in the diffuse-layer potential.

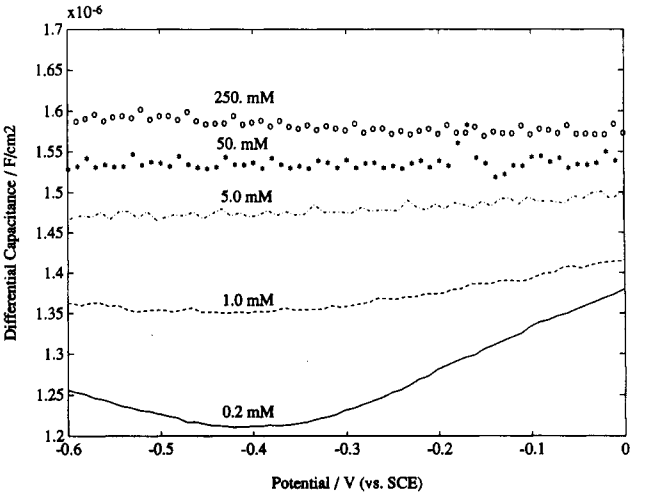


Figure 5. Differential capacitances measured at a $CH_3(CH_2)_{11}SH$ monolayer coated Au electrode as a function of the concentration of the CF_3COONa electrolyte indicated in the figure. The capacitances were calculated from cyclic voltammograms obtained at 0.5 V/s scan rate at 0.0 °C.

Comparison of Hydroxymethyl- and Methyl-Terminated Thiol Monolayers. There has been significant interest in the electrochemical properties of simple alkanethiol monolayers adsorbed onto Au electrodes.[15] Presumably, the same two-capacitor model should also describe the double layer present at Au electrodes derivatized with self-assembled alkanethiol monolayers. Figure 5 shows the capacitance of an Au electrode coated with a monolayer of dodecylmercaptan. Dodecyl mercaptan was chosen because its capacitance is approximately equal to that of a monolayer of $HO(CH_2)_{14}SH$, which should give rise to the same diffuse-layer potential. The first difference in the double-layer structure of the alkanethiol monolayer from the ω-hydroxy thiol monolayer is that the potential of zero charge as measured by the minimum of the electrode capacitance is shifted negative by approximately 300 mV. The reproducibility of this pzc determination was significantly worse for the alkyl monolayer than for the ω-hydroxy thiol monolayer derivatized electrode. The capacitance minimum was observed to vary by as much as ±0.2 V from that determined from Figure 5. Part of the variability in the pzc measured at the alkanethiol monolayers may stem from their tendency to adsorb impurities from the electrolyte. The capacitances of the alkanethiol monolayer coated Au electrodes were observed to decrease slowly with time due to the adsorption of impurities from solution. Careful cleaning of the electrochemical cell and the N_2 gas used to purge the cell dramatically improved the stability of the electrode capacitance.

The diffuse-layer potential for the dodecanethiol-derivatized Au electrode can be calculated from the capacitance data shown in Figure 5 using eq 3. Figure 6 plots the measured and calculated diffuse-layer potentials for the alkanethiol-derivatized Au electrode. As was the case for the data for the ω-hydroxy thiol coated electrode shown in Figure 2, the shape and magnitude of the calculated diffuse-layer potential are in very close agreement with those determined via the capacitance measurement.

The differences in the potentials of zero charge between the alkanethiol and ω-hydroxy thiol monolayer coated electrodes are a bit surprising. The major difference between these two monolayer-coated electrodes is their surface hydrophobicity. The alkanethiol monolayer is extremely hydrophobic, which may tend to allow the larger CF_3COO^- anions a closer approach to the electrode surface than the more highly aquated Na^+ cations. This preconcentration of the anions would tend to push the potential of zero charge negative, as is observed for these monolayers. Because we observed significant variability in the potentials of zero charge for a series of dodecyl mercaptan derivatized electrodes, the preconcentration of anions at the OHP

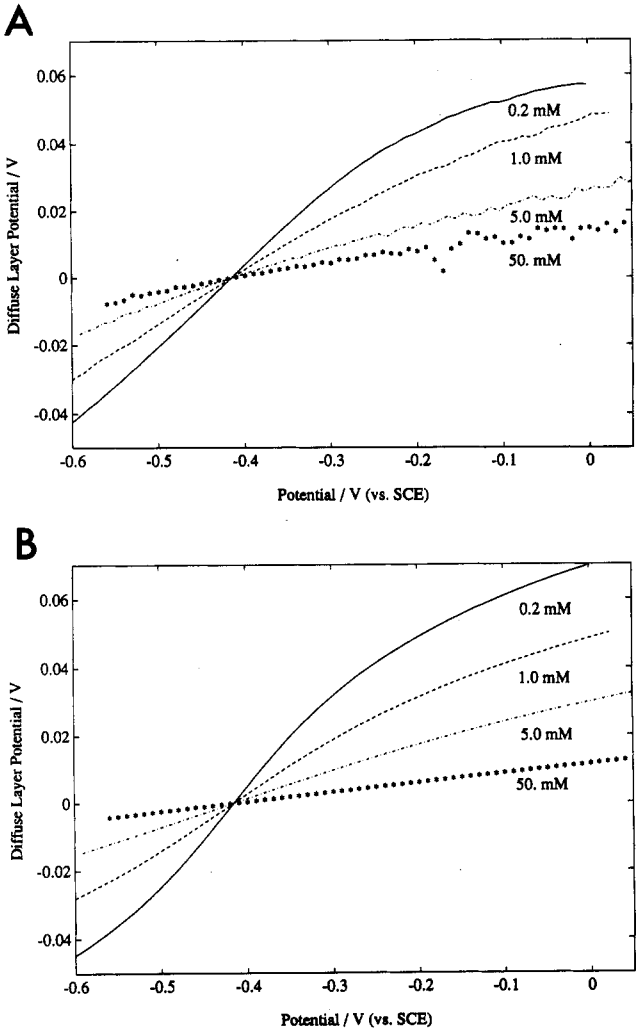


Figure 6. Electrolyte concentration dependence of the diffuse-layer potential versus the applied potential at a $CH_3(CH_2)_{11}SH$ monolayer coated Au electrode. The diffuse-layer potentials shown in part A were calculated from the capacitance data shown in Figure 5 (see text). Part B plots the diffuse-layer potentials calculated using the two-capacitor model.

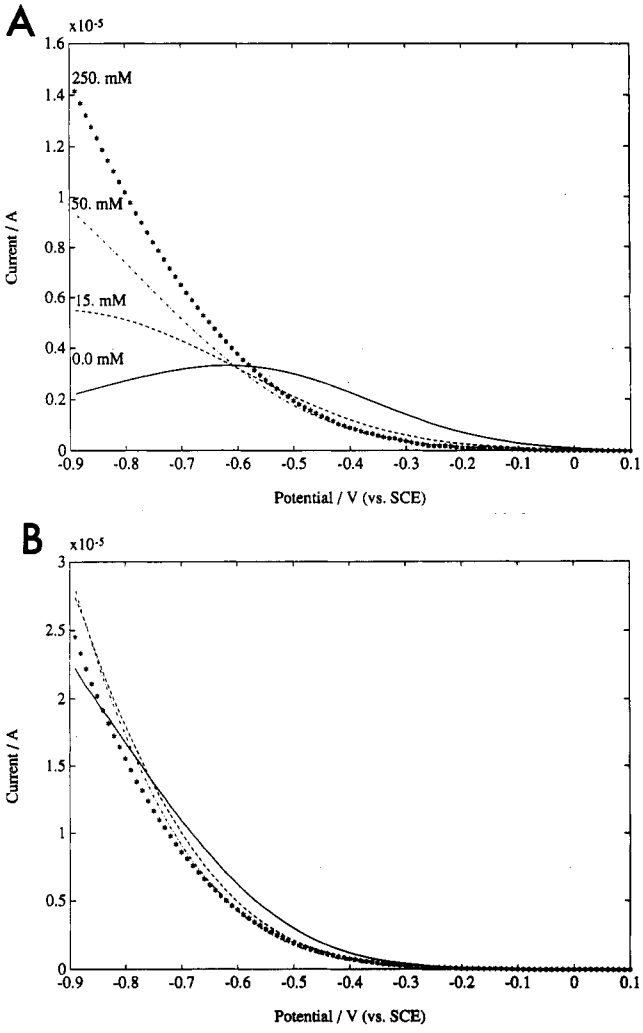


Figure 7. Correction of the heterogeneous electron-transfer rates for $Fe(CN)_6^{3-}$ reduction measured at a $CH_3(CH_2)_{11}SH$ monolayer coated Au electrode using the two-capacitor model. Part A plots the reduction currents for 2.0 mM $Fe(CN)_6^{3-}$ measured at a $CH_3(CH_2)_{11}SH$ monolayer coated Au electrode corrected for diffusion limitations via convolution techniques.[4] Part B plots these same currents after the diffuse-layer potential correction.

cannot be the only factor. The other explanation for the more negative potentials of zero charge for the dodecanethiol-coated Au electrode is that the dodecanethiol monolayer may contain defect sites which allow the specific adsorption of anions in the electrolyte onto the Au surface. The variability of the potential of zero charge could then be ascribed to difference in the number of these defect sites within the thiol monolayer.

The characterization of the double layer of alkanethiol-coated Au electrodes can be extended through heterogeneous electron-transfer measurements. The voltammetric curves for the reduction of $Fe(CN)_6^{3-}$ measured at a dodecyl mercaptan derivatized Au electrode as a function of the electrolyte concentration are shown in Figure 7A.[16] These curves were corrected for diffusion effects via convolution techniques.[4] From capacitance measurements made on this electrode and from the intersection of the voltammetric curve, a potential of zero charge of –0.55 V was measured for this particular electrode. These voltammetric currents were corrected for the diffuse-layer potential using the two-capacitor model, and the corrected currents are shown in Figure 7B. The corrected voltammetric curves are quite close to each other, lending support to the validity of the diffuse-layer calculation. As was the case for the data collected at ω-hydoxy thiol derivatized Au electrodes, the largest deviation between the corrected voltammetric currents shown in Figure 7B occurs for the data collected with no supporting electrolyte.

Kinetic Comparison of Methyl- and Hydroxymethyl-Terminated Monolayers. From the capacitance and electron-transfer measurements, the diffuse-layer potentials and potential of zero charge can be determined for both the ω-hydroxy thiol and the alkanethiol monolayer coated Au electrodes. Even when the voltammetric data are corrected for this diffuse potential, the electron-transfer kinetics measured for same redox species can vary dramatically between the ω-hydroxythiol and alkanethiol monolayer coated Au electrodes. Figure 8 shows voltammograms for the reduction of $Fe(CN)_6^{3-}$ measured using a $HO(CH_2)_{14}SH$- and a $CH_3(CH_2)_{11}SH$-derivatized Au electrode in 0.25 M CF_3COONa. Even though the dodecyl monolayer is shortened by two methylene units over the ω-hydroxy thiol monolayer, the reduction current for ferricyanide is almost 3 times smaller than that observed at the ω-hydroxy thiol monolayer. If one corrects the electron-transfer rate for the difference in the number of methylene units between the dodecanethiol and 14-hydroxytetradecane-1-thiol monolayer using the electron-transfer rate attenuation per methylene group determined previously,[4] the dodecanethiol-derivatized Au electrode is more than 20 times slower than the hydroxy thiol monolayer coated electrode. We interpret this large difference in the electron-transfer rate as primarily due to the extreme hydrophobicity of the methyl-terminated surface. In order for the $Fe(CN)_6^{3-}$ ions to come into close contact with the monolayer surface, the highly charged ion would have to become

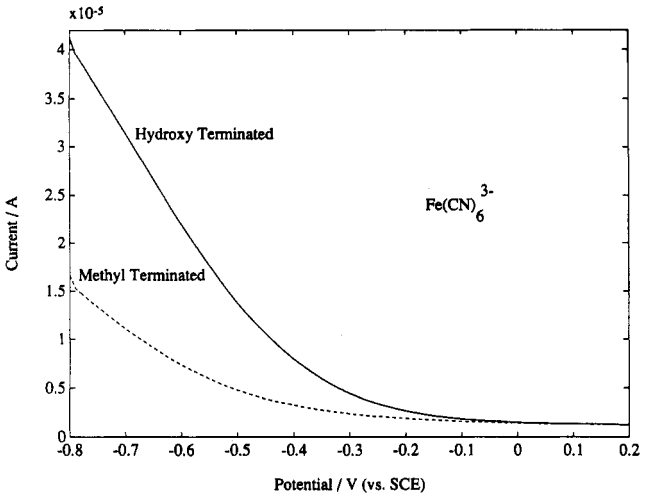


Figure 8. Comparison of the reduction rate of $Fe(CN)_6^{3-}$ measured at $HO(CH_2)_{14}SH$ and $CH_3(CH_2)_{11}SH$ monolayer coated Au electrodes. The two voltammetric currents were obtained at 0.5 V/s in a 2.5 mM $Fe(CN)_6^{3-}$, 0.25 M CF_3COONa solution held at 0.0 °C. The curves shown were not corrected for the electrode capacitance or diffusion effects.

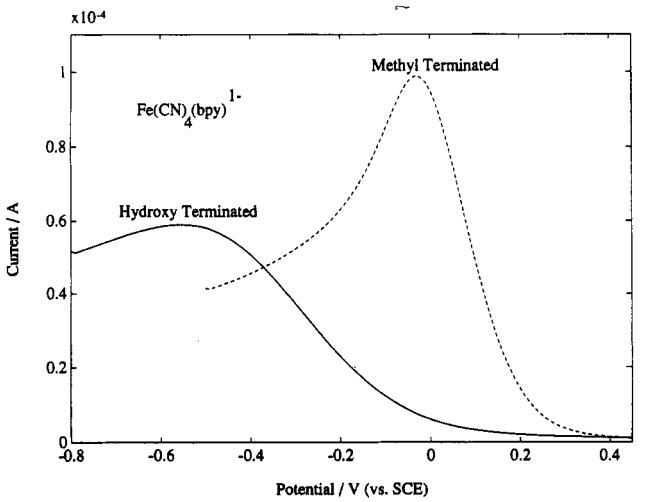


Figure 9. Comparison of the reduction rate of $Fe(bpy)(CN)_4^{1-}$ measured at $HO(CH_2)_{14}SH$ and $CH_3(CH_2)_{11}SH$ monolayer coated Au electrodes. The two voltammetric currents were obtained at 0.5 V/s in a 1.0 mM $Fe(bpy)(CN)_4^{1-}$, 0.25 M CF_3COONa solution held at 0.0 °C. The curves shown were not corrected for the electrode capacitance or diffusion effects.

partially desolvated, which would require a tremendous activation energy. The long-range electron transfer through the dodecanethiol monolayer should therefore proceed with its solvation shell largely intact. In contrast, the surface of the ω-hydroxy thiol is extremely hydrophilic, allowing the $Fe(CN)_6^{3-}$ to come into closer contact with insulating film.

The hypothesis that the solvation of the redox species gives rise to the large difference in the electron-transfer rate measured at the dodecanethiol- and 14-hydroxytetradecane-1-thiol-coated Au electrodes can be supported by comparing the reduction rates for $Fe(CN)_6^{3-}$ with a more hydrophobic, less charged electroactive molecule. Figure 9 shows the reduction currents for $Fe(bpy)(CN)_4^{1-}$ measured at the same two electrodes used to obtain the data shown in Figure 8. If one compares the reduction currents measured at the $HO(CH_2)_{14}SH$-coated Au electrode shown in Figures 8 and 9, the heterogeneous electron-transfer rate for $Fe(bpy)(CN)_4^{1-}$ is found to be significantly faster than that for $Fe(CN)_6^{3-}$, consistent with the increased intrinsic electron-transfer facility of the $Fe(bpy)(CN)_4^{1-}$ complex.[4] A comparison of the reduction rates of $Fe(bpy)(CN)_4^{1-}$ measured at the dodecanethiol- and 14-hydroxytetradecane-1-thiol derivatized Au electrode indicates an opposite trend from the reduction rate of $Fe(CN)_6^{3-}$. The $Fe(bpy)(CN)_4^{1-}$ reduction rate measured at the dodecanethiol-derivatized electrode is considerably faster than that measured at the ω-hydroxy thiol coated Au electrode, even when the rate is corrected for the difference in the length of the thiol molecules used to form the insulating monolayers. This is exactly the effect one would predict if the more hydrophobic redox molecule preferentially adsorbed at the methyl-terminated surface. In addition to the possible preconcentration of the $Fe(bpy)(CN)_4^{1-}$ at the dodecanethiol monolayer surface, the enhancement in the electron-transfer rate may also reflect the ability of the $Fe(bpy)(CN)_4^{1-}$ to penetrate defect sites within the monolayer. Such defect sites would have to be atomic in size or one might expect $Fe(CN)_6^{3-}$ to penetrate these same sites, giving rise to faster electron transfer at the dodecanethiol-coated electrode relative to the ω-hydroxy thiol coated electrode, which was not observed.

Conclusions

Although the two-capacitor model tested here is quite simplistic and contains no adjustable parameters, the model accurately predicts the diffuse-layer potential present at alkanethiol and ω-hydroxy thiol monolayer coated Au electrodes. Both capacitance measurements and electron-transfer-rate measurements support this conclusion.

A more subtle conclusion is that electron transfer to $Fe(CN)_6^{3-}$ measured at Au electrode derivatized with either dodecanethiol or 14-hydroxytetradecane-1-thiol monolayers occurs primarily through the entire thickness of the monolayer films. This conclusion stems from the close agreement between the corrected voltammetric currents as a function of the ionic strength of the electrolyte. If electron transfer to the $Fe(CN)_6^{3-}$ were taking place through a small number of defect sites within the monolayer, the diffuse-layer potential at these defect sites would be larger than that predicted by the two-capacitor model and would give rise to a larger dependence of the heterogeneous electron-transfer rate on the electrolyte concentration. The diffuse-layer studies presented here can therefore be used to strengthen our assertion that electron transfer at the ω-hydroxy thiol monolayer coated Au electrodes proceeds through the entire thickness of the monolayer. An identical argument can be made for the electron transfer to $Fe(CN)_6^{3-}$ measured at Au electrode derivatized with dodecanethiol monolayers.

Although the diffuse-layer potential is well described by the two-capacitor model, the surface hydrophobicity of the insulating monolayer can induce dramatic changes in the concentration of redox species at the insulated electrode. The apparent preconcentration of the $Fe(bpy)(CN)_4^{1-}$ at the dodecanethiol monolayer surface is a likely result of the hydrophobicity of the surface. Similarly, the much slower reduction rate measured for $Fe(CN)_6^{3-}$ at the dodecanethiol-derivatized Au electrode relative to the 14-hydroxytetradecane-1-thiol-derivatized Au electrode is also attributed to the extreme difference in the surface hydrophobicities of the two insulating films.

References and Notes

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(16) It is interesting to note that the heterogeneous electron-transfer rate for the reduction of $Fe(CN)_6^{3-}$ measured at both the dodecanethiol and 14-hydroxytetradecane-1-thiol monolayer coated Au electrodes displays a maximum at –0.6 to –0.7 V vs SCE in the absence of supporting electrolyte. Normally, the electron-transfer rate would be expected to increase dramatically with increasing overpotentials. However, at overpotentials beyond the reorganization energy of the redox molecule this dependence of the electron-transfer rate on the electrode overpotential becomes less pronounced, so the voltage dependence of the diffuse-layer potential and its effect of repelling the anionic redox molecule from the OHP prevail.